

Matchings in sparse random graphs

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Combinatorics at the Confluence

20-22 July 2026, CMU

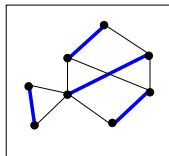
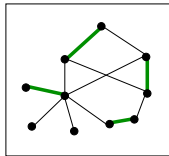
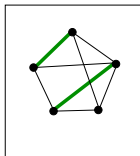
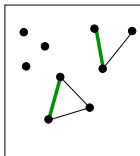
Outline of the talk

- I. Karp-Sipser algorithm
- II. Matching number of Erdős-Rényi random graph
- III. Karp-Sipser core of Erdős-Rényi random graph
- IV. Extensions
- V. Summary and open problems

Matching and perfect matching

Given a graph $G = (V, E)$,

- a **matching** $M \subset E$ is a **set of vertex-disjoint edges**
- a matching M is **perfect** if every vertex in V is incident to an edge in M



Obstacles for large matchings

- ▷ isolated vertices, cherries, odd components, ...

Sharp threshold for perfect matching

$G \sim G(n, p)$ a binomial random graph

[ERDŐS–RÉNYI 66], [BOLLOBÁS–THOMASON 1985]

Let

$$p = \frac{\ln n + c(n)}{n}$$

Then

$$\mathbb{P}[G \text{ contains a perfect matching}] \xrightarrow{n \rightarrow \infty} \begin{cases} 0 & \text{if } c(n) \rightarrow -\infty \\ 1 & \text{if } c(n) \rightarrow +\infty \end{cases}$$

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- * What about a **maximum matching** in a **sparse** random graph, when $np = c \in \mathbb{R}^+$?

Maximum matching

Given a graph $G = (V, E)$,

- a maximum matching is a matching that contains the largest number of possible edges

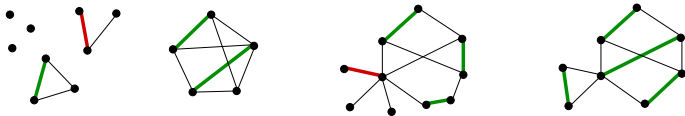
- $\nu(G) = \left| \text{a maximum matching in } G \right| = \text{matching number of } G$

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- Key observation

$\exists$ **pendent edge** $e \in E \implies \exists$ **maximum matching containing** e

Karp-Sipser algorithm

Phase 1: greedy leaf-removal process

Input: a graph $G = (V, E)$ and $M = \emptyset$

▷ while $\exists$ a **leaf**, i.e., a **vertex x of degree one**

remove it together with its unique neighbour y

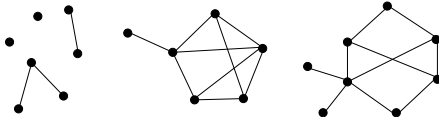
(so the vertices x, y form a **pendent edge $e = \{x, y\}$** in G)

$G \leftarrow G \setminus \{x, y\}$ and **$M \leftarrow M \cup \{e\}$**

▷ remove all isolated vertices

Output: Karp-Sipser core $G_1 = (V_1, E_1)$ and a matching M_1

G



$M = \emptyset$

Karp-Sipser algorithm

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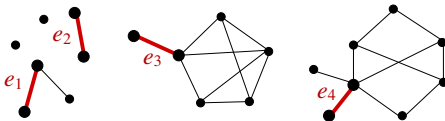
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$M = \{e_1, e_2, e_3, e_4\}$

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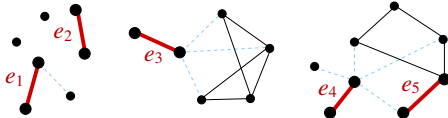
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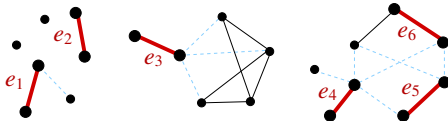
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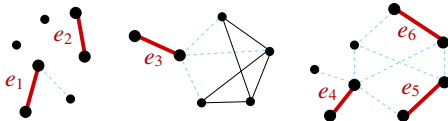
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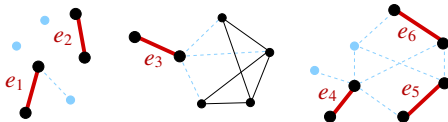
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Input: a graph $G = (V, E)$ and $M = \emptyset$

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(so the vertices x, y form a pendent edge $e = \{x, y\}$ in G)
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Karp-Sipser algorithm

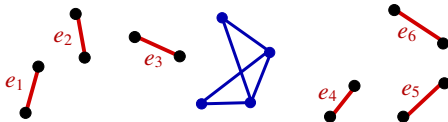
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$$M_1 = \{e_1, e_2, e_3, e_4, e_5, e_6\}$$

Karp-Sipser algorithm

Phase 2: greedy edge-removal process

Input: **Karp-Sipser core** $G_1 = (V_1, E_1)$ and a **matching** M_1

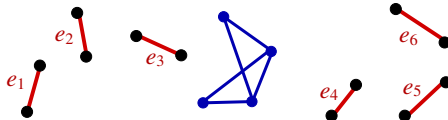
▷ while $\exists$ an edge

select **randomly an edge** $e = \{x, y\}$

$G_1 \leftarrow G_1 \setminus \{x, y\}$ and $M \leftarrow M \cup \{e\}$

Output: a matching M

Karp-Sipser core G_1



Matching $M_1 = \{e_1, e_2, e_3, e_4, e_5, e_6\}$

Karp-Sipser algorithm

Phase 2: greedy edge-removal process

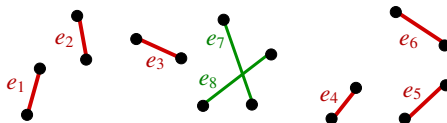
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Output: a matching M



Matching $M = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$

Part II.

Matching number of Erdős-Rényi random graph

Typical matching number

$G \sim G(n, p)$ a binomial random graph

$\nu(G)$ matching number of G

$np = c \in \mathbb{R}^+$ constant

Convergence in probability

[KARP-SIPSER 1981]

• for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\left| \frac{\nu(G)}{n/2} - \nu_c \right| > \epsilon \right] = 0$$

where $\nu_c = 1 - \frac{1}{2}(t_c + \exp(-ct_c) + ct_c \exp(-ct_c))$

and $t_c \in [0, 1]$ is the smallest root of $t = \exp(-c \exp(-ct))$

Typical matching number

$G \sim G(n, p)$ a binomial random graph

$\nu(G)$ matching number of G

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Convergence in probability & **weak law of large numbers** [KARP-SIPSER 1981]

• for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\left| \frac{\nu(G)}{n/2} - \nu_c \right| > \epsilon \right] = 0$$

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and $t_c \in [0, 1]$ is the smallest root of $t = \exp(-c \exp(-ct))$

• whp $\mathbb{E}[\nu(G)] = \nu_c n/2 + o(n)$

Central limit theorem

$G \sim G(n, p)$ a binomial random graph

$\nu(G)$ matching number of G

$np = c \in \mathbb{R}^+$ constant

Conjecture for **central limit theorem**

[ARONSON–FRIEZE–PITTEL 1998]

$$\frac{\nu(G) - \mathbb{E}[\nu(G)]}{\sqrt{\mathbb{V}[\nu(G)]}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Verified for

● $c < 1$

[Pittel 90]

Central limit theorem

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Verified for

● $c < 1$

[Pittel 90]

● $c > e$

[Kreačić 2017]

● $c > 0$

[Glasgow–Kwan–Sah–Sawhney 2025]

Local central limit theorem

$\mathbf{G} \sim \mathbf{G}(n, p)$ a binomial random graph

$\nu(\mathbf{G})$ matching number of $\mathbf{G}$

$np = c \in \mathbb{R}^+$ constant

Work in progress

[ANASTOS-ERDE-K.-PFENNINGER]

For appropriate ranges of $k \in \mathbb{N}$,

$$\mathbb{P}[\nu(\mathbf{G}) = k] = (1 + o_n(1)) \frac{1}{\sqrt{2\pi \mathbb{V}[\nu(\mathbf{G})]}} \exp\left(-\frac{(k - \mathbb{E}[\nu(\mathbf{G})])^2}{2\mathbb{V}[\nu(\mathbf{G})]}\right)$$

Part III.

Karp-Sipser core of Erdős-Rényi random graph

Typical size of Karp-Sipser core

$G \sim G(n, p)$ a binomial random graph

$np = c \in \mathbb{R}^+$ constant

$\mu(G) = |V(\text{Karp-Sipser core of } G)|$

e -phenomenon & **weak law of large numbers**

[KARP-SIPSER 1981]

• for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\left| \frac{\mu(G)}{n} - \mu_c \right| > \epsilon \right] = 0$$

where $\mu_c = (t_c - \exp(-ct_c))(1 - c \exp(-ct_c)) \cdot \mathbb{1}_{c > e}$

and $t_c \in [0, 1]$ is the smallest root of $t = \exp(-c \exp(-ct))$

• whp $\mathbb{E}[\mu(G)] = \mu_c n + o(n)$

Subcritical and critical regimes

$G \sim G(n, p)$ a binomial random graph

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$\mu(G) = |V(\text{Karp-Sipser core of } G)|$

Poisson regime

[ARONSON-FRIEZE-PITTEL 1998]

If $c < e$, then

$\mu(G)$ converges in distribution to a sum of Poisson random variables

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Sublinear regime: conjecture

[BAUER-GOLINELLI 01]

If $c = e$, then $\mu(G) = O(n^{3/5})$

Verified for

• a random graph with degrees 1, 2, 3

[BUDZINSKI-CONTAT-CURIEN 25]

Local central limit theorem

$G \sim G(n, p)$ a binomial random graph

$np \geq C$ for large constant $C > e$

$\mu(G) = |V(\text{Karp-Sipser core of } G)|$

[ANASTOS-ERDE-K.-PFENNINGER 26+]

For appropriate ranges of $k \in \mathbb{N}$,

$$\mathbb{P}[\mu(G) = k] = (1 + o_n(1)) \frac{1}{\sqrt{2\pi \mathbb{V}[\mu(G)]}} \exp\left(-\frac{(k - \mathbb{E}[\mu(G)])^2}{2\mathbb{V}[\mu(G)]}\right)$$

Part IV.

Extensions

Universality of matching number

$(G_n)_{n \in \mathbb{N}}$ a sequence of random graphs
whose **Benjamini-Schramm local weak limit** is
Galton-Watson tree with $\text{Po}(c)$ for constant $c \in \mathbb{R}^+$

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 $\nu(G_n)$ matching number of G_n

[LELARGE 2013], [BORDENAVE–LELARGE–SALEZ 2013]

$$\lim_{n \rightarrow \infty} \frac{\nu(G_n)}{n/2} = \nu_c = 1 - \max_{\alpha \in [0,1]} \Phi_c(\alpha) \quad \text{in probability}$$

where $\Phi_c : [0, 1] \rightarrow [0, 1]$, $\alpha \mapsto \Phi_c(\alpha)$, is **Bethe free entropy** defined as

$$\Phi_c(\alpha) := -1 + \exp \left(-c \exp(-c(1 - \alpha)) \right) + (1 + c(1 - \alpha) \exp(-c(1 - \alpha)))$$

- ▷ true in more general setting
- ▷ recursive distributional equation

Universality of size of Karp-Sipser core

$(G_n)_{n \in \mathbb{N}}$ a sequence of random graphs
whose **Benamini-Schramm local weak limit** is
Galton-Watson tree with $\text{Po}(c)$ for constant $c \in \mathbb{R}^+$

$$\mu(G_n) = |V(\text{Karp-Sipser core of } G_n)|$$

Convergence in probability

[CHATTERJEE–LEE–ZHU 2025+], [LIU 2025+]

$$\lim_{n \rightarrow \infty} \frac{\mu(G_n)}{n} = \mu_c \quad \text{in probability}$$

- ▷ true in configuration model with arbitrary degree distribution
- ▷ Warning/Belief propagation message passing algorithms

Random subgraphs

Given a graph G and $p \in (0, 1)$,

G_p = a **random subgraph of G** with **edge-percolation probability p**
= a random subgraph of G obtained by retaining each edge of G independently with probability p

Random subgraphs

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= a random subgraph of G obtained by retaining each edge of G independently with probability p

For example,

K_p^n = $G(n, p)$
= a random subgraph of the n -vertex complete graph K^n with edge-percolation probability p

Q_p^d = a random subgraph of the d -dimensional binary hypercube Q^d with edge-percolation probability p

perfect matching in random subgraphs

[ERDŐS–RÉNYI 66], [BOLLOBÁS–THOMASON 1985]

$p = \frac{\ln n}{n}$ is a sharp threshold for a perfect matching in K_p^n

perfect matching in random subgraphs

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$p = \frac{\ln n}{n}$ is a sharp threshold for a perfect matching in K_p^n

[BOLLOBÁS 1990]

$p = \frac{1}{2}$ is a sharp threshold for a perfect matching in $\mathcal{Q}_p^d$

Almost perfect matching in random subgraphs

[ERDŐS–RÉNYI 66], [BOLLOBÁS–THOMASON 1985]

$p = \frac{\ln n}{n}$ is a sharp threshold for a perfect matching in K_p^n

[BOLLOBÁS 1990]

$p = \frac{1}{2}$ is a sharp threshold for a perfect matching in $\mathcal{Q}_p^d$

[DISKIN–ERDE–K.–KRIVELEVICH 2026]

If $dp \geq C$ for large constant $C \in \mathbb{R}^+$, then whp

$$(1 - o_C(1)) 2^{d-1} \leq \nu(\mathcal{Q}_p^d) \leq 2^{d-1}$$

Matching number

G **approx. d -regular** n -vertex graph with **$\deg(v) = (1 + o_d(1))d$**

Matching number

G **approx. d -regular** n -vertex graph with $\deg(v) = (1 + o_d(1))d$

G_p random subgraph of G with edge-percolation probability p

$dp = c \in \mathbb{R}^+$ constant and $d = \omega(1)$

Local weak limit

[DISKIN-K.-LICHEV 2025]

Benjamini-Schramm local weak limit of G_p is $\text{Po}(c)$ Galton-Watson tree

Matching number

G **approx. d -regular** n -vertex graph with $\deg(v) = (1 + o_d(1))d$

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Consequence: convergence in probability

[DISKIN–K.–LICHEV 2025]

$$\lim_{n \rightarrow \infty} \frac{\nu(G_p)}{n/2} = \nu_c = 1 - \max_{\alpha \in [0,1]} \Phi_c(\alpha) \quad \text{in probability}$$

$$\lim_{n \rightarrow \infty} \frac{\mu(G_n)}{n} = \mu_c \quad \text{in probability}$$

Part V.

Summary and open problems

Matching number of ER random graph

$G \sim G(n, p)$ a binomial random graph

$\nu(G)$ matching number of G

$np = c \in \mathbb{R}^+$ constant

Convergence in probability

[KARP-SIPSER 81], [BORDENAVE–LELARGE–SALEZ 2013]

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Central limit theorem

[PITTEL 90], [KREAČIĆ 2017], [GLASGOW-KWAN-SAH-SAWHNEY 2025]

$$\frac{\nu(G) - \mathbb{E}[\nu(G)]}{\sqrt{\mathbb{V}[\nu(G)]}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Local central limit theorem

[ANASTOS-ERDE-K.-PFENNINGER 26+]

$$\mathbb{P}[\mu(G) = k] = (1 + o_n(1)) \frac{1}{\sqrt{2\pi \mathbb{V}[\mu(G)]}} \exp\left(-\frac{(k - \mathbb{E}[\mu(G)])^2}{2\mathbb{V}[\mu(G)]}\right)$$

Matching number of random subgraph of Q^d

Q_p^d random subgraph of Q^d with edge-percolation probability p

$\nu(Q_p^d)$ matching number of Q_p^d

$dp = c \in \mathbb{R}^+$ constant and $d = \omega(1)$

Convergence in probability

[DISKIN-K.-LICHEV 2025]

$$\lim_{d \rightarrow \infty} \frac{\nu(Q_p^d)}{2^{d-1}} = \nu_c = 1 - \max_{\alpha \in [0,1]} \Phi_c(\alpha) \quad \text{in probability}$$

Matching number of random subgraph of Q^d

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Convergence in probability

[DISKIN–K.–LICHEV 2025]

$$\lim_{d \rightarrow \infty} \frac{\nu(Q_p^d)}{2^{d-1}} = \nu_c = 1 - \max_{\alpha \in [0,1]} \Phi_c(\alpha) \quad \text{in probability}$$

Stronger limit laws for $\nu(Q_p^d)$

Q. Central limit theorem ?

Q. Local central limit theorem ?

Karp-Sipser of ER random graph

$G \sim G(n, p)$ a binomial random graph

$$\mu(G) = |V(\text{Karp-Sipser core of } G)|$$

Convergence in probability

[KARP-SIPSER 81]

If $np = c \in \mathbb{R}^+$ constant, then

$$\lim_{n \rightarrow \infty} \frac{\mu(G)}{n} = \mu_c$$

in probability

Karp-Sipser of ER random graph

$G \sim G(n, p)$ a binomial random graph

$$\mu(G) = |V(\text{Karp-Sipser core of } G)|$$

Convergence in probability

[KARP-SIPSER 81]

If $np = c \in \mathbb{R}^+$ constant, then

$$\lim_{n \rightarrow \infty} \frac{\mu(G)}{n} = \mu_c \quad \text{in probability}$$

Local central limit theorem

[ANASTOS-ERDE-K.-PFENNINGER 26+]

If $np \geq C$ for large constant $C \in \mathbb{R}^+$, then

$$\mathbb{P}[\mu(G) = k] = (1 + o_n(1)) \frac{1}{\sqrt{2\pi \mathbb{V}[\mu(G)]}} \exp\left(-\frac{(k - \mathbb{E}[\mu(G)])^2}{2\mathbb{V}[\mu(G)]}\right)$$

Karp-Sipser of random subgraph of hypercube

$\mathcal{Q}_p^d$ random subgraph of Q^d with edge-percolation probability p

$dp = c \in \mathbb{R}^+$ constant and $d = \omega(1)$

$$\mu(\mathcal{Q}_p^d) = |V(\text{Karp-Sipser core of } \mathcal{Q}_p^d)|$$

Convergence in probability

[DISKIN–K.–LICHEV 2025]

$$\lim_{d \rightarrow \infty} \frac{\mu(\mathcal{Q}_p^d)}{2^d} = \mu_c$$

in probability

Karp-Sipser of random subgraph of hypercube

$\mathcal{Q}_p^d$ random subgraph of Q^d with edge-percolation probability p

$dp = c \in \mathbb{R}^+$ constant and $d = \omega(1)$

$$\mu(\mathcal{Q}_p^d) = |V(\text{Karp-Sipser core of } \mathcal{Q}_p^d)|$$

Convergence in probability

[DISKIN–K.–LICHEV 2025]

$$\lim_{d \rightarrow \infty} \frac{\mu(\mathcal{Q}_p^d)}{2^d} = \mu_c$$

in probability

Stronger limit laws for $\mu(\mathcal{Q}_p^d)$

Q. Central limit theorem ?

Q. Local central limit theorem ?